

Git and Node.js are already installed

```
Windows PowerShell
PS C:\Users\diurno> node -v
v24.21.0
PS C:\Users\diurno> git --version
git version 2.55.0.windows.5
PS C:\Users\diurno> |
```

Use the next command for install global Node modules:

`npm install -g bunyan gulp@4`

```
Windows PowerShell
PS C:\Users\diurno> npm install -g bunyan gulp@4
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain widely publicized security vulnerabilities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorbitant rates) by contacting i@izs.me
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if you want a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerful.
npm warn deprecated urix@0.1.0: Please see https://github.com/lydell/urix#deprecated
npm warn deprecated source-map-url@0.4.1: See https://github.com/lydell/source-map-url#deprecated
npm warn deprecated resolve-url@0.2.1: https://github.com/lydell/resolve-url#deprecated
npm warn deprecated source-map-resolve@0.5.3: See https://github.com/lydell/source-map-resolve#deprecated
npm warn deprecated rimraf@2.4.5: Rimraf versions prior to v4 are no longer supported
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if you want a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerful.
npm warn deprecated glob@6.0.4: Old versions of glob are not supported, and contain widely publicized security vulnerabilities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorbitant rates) by contacting i@izs.me
added 360 packages in 16s

18 packages are looking for funding
  run `npm fund` for details
npm warn install-scripts 2 packages have install scripts not yet covered by allowScripts:
npm warn install-scripts dtrace-provider@0.8.8 (install: node-gyp rebuild || node suppress-error.js)
npm warn install-scripts es5-ext@0.10.64 (postinstall: node -e "try{require('./_postinstall')}}catch(e){}" || exit 0)
npm warn install-scripts Run `npm install -g --allow-scripts=dtrace-provider,es5-ext` to allow these scripts once, or `npm config set allow-scripts=dtrace-provider,es5-ext --location=user` to allow them for all global installs.
npm notice
npm notice New major version of npm available! 11.19.0 -> 12.0.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.0.2
npm notice To update run: npm install -g npm@12.0.2
npm notice
PS C:\Users\diurno> |
```

Create a folder, in my case this is the path:

“C:\Users\diurno\Desktop\Libro JS”

Clone the tutorial server into the created folder:

`cd /Libro JS`

`git clone https://github.com/javascript-tutorial/server`

`git clone https://github.com/javascript-tutorial/engine server/modules/engine`

```

PS C:\Users\diurno\Desktop\Libro JS> git clone https://github.com/javascript-tutorial/server
Cloning into 'server'...
remote: Enumerating objects: 3245, done.
remote: Counting objects: 100% (218/218), done.
remote: Compressing objects: 100% (58/58), done.
remote: Total 3245 (delta 185), reused 160 (delta 160), pack-reused 3027 (from 2)
Receiving objects: 100% (3245/3245), 3.66 MiB | 4.71 MiB/s, done.
Resolving deltas: 100% (1446/1446), done.
PS C:\Users\diurno\Desktop\Libro JS> git clone https://github.com/javascript-tutorial/engine server/modules/engine
Cloning into 'server/modules/engine'...
remote: Enumerating objects: 4117, done.
remote: Counting objects: 100% (423/423), done.
remote: Compressing objects: 100% (267/267), done.
remote: Total 4117 (delta 211), reused 230 (delta 148), pack-reused 3694 (from 2)
Receiving objects: 100% (4117/4117), 3.21 MiB | 12.08 MiB/s, done.
Resolving deltas: 100% (2465/2465), done.
PS C:\Users\diurno\Desktop\Libro JS> |

```

Use the next commands for clone other repository in a child folder

cd server/repo

git clone https://github.com/javascript-tutorial/es.javascript.info

```

PS C:\Users\diurno\Desktop\Libro JS> cd .\server\repo\
PS C:\Users\diurno\Desktop\Libro JS\server\repo> git clone https://github.com/javascript-tutorial/es.javascript.info
Cloning into 'es.javascript.info'...
remote: Enumerating objects: 81384, done.
remote: Counting objects: 100% (611/611), done.
remote: Compressing objects: 100% (285/285), done.
remote: Total 81384 (delta 405), reused 405 (delta 318), pack-reused 80773 (from 3)
Receiving objects: 100% (81384/81384), 717.16 MiB | 29.58 MiB/s, done.
Resolving deltas: 100% (48173/48173), done.
PS C:\Users\diurno\Desktop\Libro JS\server\repo> |

```

Go to the previous folder and install npm

cd ..

npm install

```

PS C:\Users\diurno\Desktop\Libro JS\server\repo> cd ..
PS C:\Users\diurno\Desktop\Libro JS\server> npm install
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru-cache if
you want a good and tested way to coalesce async requests by a key value, which is much more comprehensive and powerfu
l.
npm warn deprecated source-map-url@0.4.1: See https://github.com/lydell/source-map-url#deprecated
npm warn deprecated rimraf@2.4.5: Rimraf versions prior to v4 are no longer supported
npm warn deprecated urix@0.1.0: Please see https://github.com/lydell/urix#deprecated
npm warn deprecated glob@6.0.4: Old versions of glob are not supported, and contain widely publicized security vulnerabi
lities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorb
itant rates) by contacting i@izs.me
npm warn deprecated resolve-url@0.2.1: https://github.com/lydell/resolve-url#deprecated
npm warn deprecated source-map-resolve@0.5.3: See https://github.com/lydell/source-map-resolve#deprecated
npm warn deprecated har-validator@5.1.5: this library is no longer supported
npm warn deprecated glob@7.1.3: Old versions of glob are not supported, and contain widely publicized security vulnerabi
lities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exorb
itant rates) by contacting i@izs.me
npm warn deprecated request-promise@4.2.6: request-promise has been deprecated because it extends the now deprecated req
uest package, see https://github.com/request/request/issues/3142
npm warn deprecated raw-body@1.1.7: No longer maintained. Please upgrade to a stable version.
npm warn deprecated @babel/plugin-proposal-object-rest-spread@7.20.7: This proposal has been merged to the ECMAScript st
andard and thus this plugin is no longer maintained. Please use @babel/plugin-transform-object-rest-spread instead.
npm warn deprecated koa-router@7.4.0: Please use @koa/router instead, starting from v9!
npm warn deprecated debug@3.2.6: Debug versions >=3.2.0 <3.2.7 || >=4 <4.3.1 have a low-severity ReDos regression when u
sed in a Node.js environment. It is recommended you upgrade to 3.2.7 or 4.3.1. (https://github.com/visionmedia/debug/iss
ues/797)
npm warn deprecated node-uuid@1.4.8: Use uuid module instead
npm warn deprecated request@2.88.2: request has been deprecated, see https://github.com/request/request/issues/3142
npm warn deprecated uuid@3.4.0: uuid@10 and below is no longer supported. For ESM codebases, update to uuid@latest. Fo
r CommonJS codebases, use uuid@11 (but be aware this version will likely be deprecated in 2028).
npm warn deprecated gm@1.25.1: The gm module has been sunset. Please migrate to an alternative. https://github.com/aheck
mann/gm?tab=readme-ov-file#2025-02-24-this-project-is-not-maintained
npm warn deprecated uglify-es@3.3.10: support for ECMAScript is superseded by 'uglify-js' as of v3.13.0
npm warn gitignore-fallback No .npmignore file found, using .gitignore for file exclusion. Consider creating a .npmignor
e file to explicitly control published files.
npm warn deprecated uuid@8.3.2: uuid@10 and below is no longer supported. For ESM codebases, update to uuid@latest. Fo
r CommonJS codebases, use uuid@11 (but be aware this version will likely be deprecated in 2028).
npm warn deprecated glob@10.5.0: Old versions of glob are not supported, and contain widely publicized security vulnerab
ilities, which have been fixed in the current version. Please update. Support for old versions may be purchased (at exor
bitant rates) by contacting i@izs.me
npm warn deprecated core-js@2.6.12: core-js@<3.23.3 is no longer maintained and not recommended for usage due to the num
ber of issues. Because of the V8 engine whims, feature detection in old core-js versions could cause a slowdown up to 10
0x even if nothing is polyfilled. Some versions have web compatibility issues. Please, upgrade your dependencies to the
actual version of core-js.

added 1323 packages, and audited 1324 packages in 51s

176 packages are looking for funding
  run `npm fund` for details

66 vulnerabilities (2 low, 23 moderate, 15 high, 26 critical)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
npm warn install-scripts 3 packages have install scripts not yet covered by allowScripts:
npm warn install-scripts core-js@2.6.12 (postinstall: node -e "try{require('./postinstall')}catch(e){}")
npm warn install-scripts dtrace-provider@0.8.8 (install: node-gyp rebuild || node suppress-error.js)
npm warn install-scripts es5-ext@0.10.64 (postinstall: node -e "try{require('./_postinstall')}catch(e){}" || exit 0)
npm warn install-scripts
npm warn install-scripts Run `npm install-scripts ls` to review, or `npm install-scripts approve <pkg>` to allow.
PS C:\Users\diurno\Desktop\Libro JS\server> |

```

Launch the server located in server folder using:

./edit es

```

PS C:\Users\diurno\Desktop\Libro JS\server> ./edit es

C:\Users\diurno\Desktop\Libro JS\server>call gulp edit | bunyan
[10:35:24] Using gulpfile ~\Desktop\Libro JS\server\gulpfile.js
[10:35:24] Starting 'edit'...
[10:35:24] Starting 'webpack'...
[10:35:24] Starting 'engine:koa:tutorial:importWatch'...
[10:35:24] Starting 'livereload'...
[10:35:24] Starting 'server'...
Livereload server listening
[2026-09-16T08:35:25.202Z] INFO: application/4264 on ALUMNO03_AULAB3: Boot complete
[2026-09-16T08:35:25.229Z] INFO: tutorialImporter/4264 on ALUMNO03_AULAB3: sync PLNKR_ENABLED=
[2026-09-16T08:35:25.229Z] INFO: tutorialImporter/4264 on ALUMNO03_AULAB3: sync C:\Users\diurno\Desktop\Libro JS\server\repo\es.javascript.info\1-js
C:\Users\diurno\Desktop\Libro JS\server\repo\es.javascript.info\1-js C:\Users\diurno\Desktop\Libro JS\server\repo\es.javascript.info
[2026-09-16T08:35:25.230Z] INFO: tutorialImporter/4264 on ALUMNO03_AULAB3: syncFolder C:\Users\diurno\Desktop\Libro JS\server\repo\es.javascript.info\1-js
[2026-09-16T08:35:25.230Z] INFO: tutorialImporter/4264 on ALUMNO03_AULAB3: syncFolder C:\Users\diurno\Desktop\Libro JS\server\repo\es.javascript.info\1-js\01-getting-started
[2026-09-16T08:35:25.230Z] INFO: tutorialImporter/4264 on ALUMNO03_AULAB3: syncArticle C:\Users\diurno\Desktop\Libro JS\server\repo\es.javascript.info\1-js\01-getting-started\1-intro
[2026-09-16T08:35:25.246Z] INFO: application/4264 on ALUMNO03_AULAB3: Server is listening localhost:3000
[10:35:25] Finished 'server' after 817 ms
[2026-09-16T08:35:25.323Z] INFO: tutorialImporter/4264 on ALUMNO03_AULAB3: importImage C:\Users\diurno\Desktop\Libro JS\server\repo\es.javascript.info\1-js\01-getting-started\1-intro\limitations.svg to C:\Users\diurno\Desktop\Libro JS\server\public\en\article\intro
[2026-09-16T08:35:25.325Z] INFO: tutorialImporter/4264 on ALUMNO03_AULAB3: syncArticle C:\Users\diurno\Desktop\Libro JS\server\repo\es.javascript.info\1-js\01-getting-started\2-manuals-specifications
[2026-09-16T08:35:25.466Z] INFO: tutorialImporter/4264 on ALUMNO03_AULAB3: syncArticle C:\Users\diurno\Desktop\Libro JS\server\repo\es.javascript.info\1-js\01-getting-started\3-code-editors
[2026-09-16T08:35:25.530Z] INFO: tutorialImporter/4264 on ALUMNO03_AULAB3: syncArticle C:\Users\diurno\Desktop\Libro JS\server\repo\es.javascript.info\1-js\01-getting-started\4-devtools
[2026-09-16T08:35:25.609Z] INFO: tutorialImporter/4264 on ALUMNO03_AULAB3: importImage C:\Users\diurno\Desktop\Libro JS\server\repo\es.javascript.info\1-js\01-getting-started\4-devtools\safari.png to C:\Users\diurno\Desktop\Libro JS\server

```

In the browser go to:

<http://localhost:3000>

The Modern JavaScript Tutorial

PART 1

El lenguaje JavaScript

Aquí aprenderemos JavaScript, empezando desde cero y llegaremos hasta conceptos avanzados como POO.

Nos concentraremos en el lenguaje mismo con el mínimo de notas específicas del entorno.

Una introducción

- 1.1

Una introducción a JavaScript
- 1.2

Manuales y especificaciones
- 1.3

Editores de Código
- 1.4

Consola de desarrollador

Fundamentos de JavaScript

- 2.1

¡Hola, mundo!
- 2.2

Estructura del código
- 2.3

El modo moderno, "use strict"
- 2.4

Variables
- 2.5

Tipos de datos
- 2.6

Interacción: alert, prompt, confirm
- 2.7

Conversiones de Tipos
- 2.8

Operadores básicos, matemáticas
- 2.9

Comparaciones
- 2.10

Ejecución condicional: if, '?'
- 2.11

Operadores Lógicos
- 2.12

Operador Nullish Coalescing '??'
- 2.13

Bucles: while y for
- 2.14

La sentencia "switch"
- 2.15

Funciones
- 2.16

Expresiones de función
- 2.17

Funciones Flecha, lo básico
- 2.18

Especiales JavaScript

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- 3.1

Debugging en el navegador
- 3.2

Estilo de codificación
- 3.3

Comentarios
- 3.4

Código ninja
- 3.5

Automatización de pruebas con Mocha
- 3.6

Polyfills y transpiladores

Objetos: lo básico

- 4.1

Objetos
- 4.2

Referencias de objetos y copia
- 4.3

Recolección de basura
- 4.4

Métodos de objeto, "this"
- 4.5

Constructor, operador "new"
- 4.6

Encadenamiento opcional '?.'
- 4.7

Tipo Symbol
- 4.8

Conversión de objeto a valor primitivo

Tipos de datos

- 5.1

Métodos en tipos primitivos
- 5.2

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- 5.3

Strings
- 5.4

Arrays
- 5.5

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- 5.6

Iterables
- 5.7

Map y Set
- 5.8

WeakMap y WeakSet
- 5.9

Object.keys, values, entries
- 5.10

Asignación destructurante
- 5.11

Fecha y Hora
- 5.12

Métodos JSON, toJSON

Trabajo avanzado con funciones

- 6.1

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- 6.2

Parámetros Rest y spread Spread
- 6.3

Objeto Global
- 6.4

Función como objeto, MFE
- 6.5

Decoradores
- 6.6

Decoradores y redirecciones, call/apply